BOARD PARTITIONS FOR A 6×7 BOARD WITH 6 BLOCKERS

This table is a resource for the paper "Towards a Classification of Blocked Rectangular Grids and an Application to Finite Tiling Problems" by Noah Jensen and Stephanie Treneer. The example that generates this table is that of a game board composed of 6×7 square cells and with 6 blockers placed on the board. This table lists a board partition, defined in the referenced paper, K the subgroup of $\langle H, V\rangle$, the symmetry group of a rectangle, under which the board partition remains invariant, and each index $[\langle H, V\rangle : K]$. There are 115 entries.

Board partitions for a 6×7 board with 6 blockers

Board Partition	K	$[\langle H, V\rangle : K]$
(0,0,0,0,3,3,)	$\langle H, V\rangle$	1
(1,0,0,0,2,3,)	$\langle e\rangle$	4
(1,0,0,0,3,2,)	$\langle e\rangle$	4
(1,0,0,1,2,2,)	$\langle H\rangle$	2
(1,0,0,1,3,1,)	$\langle e\rangle$	4
(1,0,1,0,2,2,)	$\langle R_{180}\rangle$	2
(1,0,1,0,3,1,)	$\langle e\rangle$	4
(1,1,0,0,1,3,)	$\langle V\rangle$	2
(1,1,0,0,2,2,)	$\langle V\rangle$	2
(1,1,0,0,3,1,)	$\langle V\rangle$	2
(1,1,1,0,0,3,)	$\langle e\rangle$	4
(1,1,1,0,1,2,)	$\langle e\rangle$	4
(1,1,1,0,2,1,)	$\langle e\rangle$	4
(1,1,1,0,3,0,)	$\langle e\rangle$	4
(1,1,1,1,1,1,)	$\langle H, V\rangle$	1
(1,1,1,1,2,0,)	$\langle V\rangle$	2
(2,0,0,0,1,3,)	$\langle e\rangle$	4
(2,0,0,0,2,2,)	$\langle e\rangle$	4
(2,0,0,0,3,1,)	$\langle e\rangle$	4
(2,0,0,1,0,3,)	$\langle e\rangle$	4
(2,0,0,1,1,2,)	$\langle e\rangle$	4
(2,0,0,1,2,1,)	$\langle e\rangle$	4
(2,0,0,1,3,0,)	$\langle e\rangle$	4
(2,0,0,2,1,1,)	$\langle H\rangle$	2
(2,0,0,2,2,0,)	$\langle e\rangle$	4
(2,0,1,0,0,3,)	$\langle e\rangle$	4
(2,0,1,0,1,2,)	$\langle e\rangle$	4
(2,0,1,0,2,1,)	$\langle e\rangle$	4
(2,0,1,0,3,0,)	$\langle e\rangle$	4
(2,0,1,1,0,2,)	$\langle e\rangle$	4
(2,0,1,1,1,1,)	$\langle e\rangle$	4
(2,0,1,1,2,0,)	$\langle e\rangle$	4

Board Partition	K	$[\langle H,V\rangle : K]$
(2,0,2,0,1,1,)	$\langle R_{180}\rangle$	2
(2,0,2,0,2,0,)	$\langle e\rangle$	4
(2,1,0,0,0,3,)	$\langle e\rangle$	4
(2,1,0,0,1,2,)	$\langle e\rangle$	4
(2,1,0,0,2,1,)	$\langle e\rangle$	4
(2,1,0,0,3,0,)	$\langle e\rangle$	4
(2,1,0,1,0,2,)	$\langle e\rangle$	4
(2,1,0,1,1,1,)	$\langle e\rangle$	4
(2,1,0,1,2,0,)	$\langle e\rangle$	4
(2,1,0,2,0,1,)	$\langle e\rangle$	4
(2,1,0,2,1,0,)	$\langle e\rangle$	4
(2,1,1,0,0,2,)	$\langle e\rangle$	4
(2,1,1,0,1,1,)	$\langle e\rangle$	4
(2,1,1,0,2,0,)	$\langle e\rangle$	4
(2,1,1,1,0,1,)	$\langle e\rangle$	4
(2,1,1,1,1,0,)	$\langle e\rangle$	4
(2,1,1,2,0,0,)	$\langle H\rangle$	2
(2,1,2,0,0,1,)	$\langle e\rangle$	4
(2,1,2,0,1,0,)	$\langle e\rangle$	4
(2,1,2,1,0,0,)	$\langle R_{180}\rangle$	2
(2,2,0,0,0,2,)	$\langle V\rangle$	2
(2,2,0,0,1,1,)	$\langle V\rangle$	2
(2,2,0,0,2,0,)	$\langle V\rangle$	2
(2,2,1,0,0,1,)	$\langle e\rangle$	4
(2,2,1,0,1,0,)	$\langle e\rangle$	4
(2,2,1,1,0,0,)	$\langle V\rangle$	2
(2,2,2,0,0,0,)	$\langle e\rangle$	4
(3,0,0,0,0,3,)	$\langle e\rangle$	4
(3,0,0,0,1,2,)	$\langle e\rangle$	4
(3,0,0,0,2,1,)	$\langle e\rangle$	4
(3,0,0,0,3,0,)	$\langle e\rangle$	4
(3,0,0,1,0,2,)	$\langle e\rangle$	4
(3,0,0,1,1,1,)	$\langle e\rangle$	4
(3,0,0,1,2,0,)	$\langle e\rangle$	4
(3,0,0,2,0,1,)	$\langle e\rangle$	4
(3,0,0,2,1,0,)	$\langle e\rangle$	4
(3,0,0,3,0,0,)	$\langle H\rangle$	2
(3,0,1,0,0,2,)	$\langle e\rangle$	4
(3,0,1,0,1,1,)	$\langle e\rangle$	4
(3,0,1,0,2,0,)	$\langle e\rangle$	4
(3,0,1,1,0,1,)	$\langle e\rangle$	4
(3,0,1,1,1,0,)	$\langle e\rangle$	4
(3,0,1,2,0,0,)	$\langle e\rangle$	4

Board Partition	K	$[\langle H,V\rangle : K]$
(3,0,2,0,0,1,)	$\langle e\rangle$	4
(3,0,2,0,1,0,)	$\langle e\rangle$	4
(3,0,2,1,0,0,)	$\langle e\rangle$	4
(3,0,3,0,0,0,)	$\langle R_{180}\rangle$	2
(3,1,0,0,0,2,)	$\langle e\rangle$	4
(3,1,0,0,1,1,)	$\langle e\rangle$	4
(3,1,0,0,2,0,)	$\langle e\rangle$	4
(3,1,0,1,0,1,)	$\langle e\rangle$	4
(3,1,0,1,1,0,)	$\langle e\rangle$	4
(3,1,0,2,0,0,)	$\langle e\rangle$	4
(3,1,1,0,0,1,)	$\langle e\rangle$	4
(3,1,1,0,1,0,)	$\langle e\rangle$	4
(3,1,1,1,0,0,)	$\langle e\rangle$	4
(3,1,2,0,0,0,)	$\langle e\rangle$	4
(3,2,0,0,0,1,)	$\langle e\rangle$	4
(3,2,0,0,1,0,)	$\langle e\rangle$	4
(3,2,0,1,0,0,)	$\langle e\rangle$	4
(3,2,1,0,0,0,)	$\langle e\rangle$	4
(3,3,0,0,0,0,)	$\langle V\rangle$	2
(4,0,0,0,0,2,)	$\langle e\rangle$	4
(4,0,0,0,1,1,)	$\langle e\rangle$	4
(4,0,0,0,2,0,)	$\langle e\rangle$	4
(4,0,0,1,0,1,)	$\langle e\rangle$	4
(4,0,0,1,1,0,)	$\langle e\rangle$	4
(4,0,0,2,0,0,)	$\langle e\rangle$	4
(4,0,1,0,0,1,)	$\langle e\rangle$	4
(4,0,1,0,1,0,)	$\langle e\rangle$	4
(4,0,1,1,0,0,)	$\langle e\rangle$	4
(4,0,2,0,0,0,)	$\langle e\rangle$	4
(4,1,0,0,0,1,)	$\langle e\rangle$	4
(4,1,0,0,1,0,)	$\langle e\rangle$	4
(4,1,0,1,0,0,)	$\langle e\rangle$	4
(4,1,1,0,0,0,)	$\langle e\rangle$	4
(4,2,0,0,0,0,)	$\langle e\rangle$	4
(5,0,0,0,0,1,)	$\langle e\rangle$	4
(5,0,0,0,1,0,)	$\langle e\rangle$	4
(5,0,0,1,0,0,)	$\langle e\rangle$	4
(5,0,1,0,0,0,)	$\langle e\rangle$	4
(5,1,0,0,0,0,)	$\langle e\rangle$	4
(6,0,0,0,0,0,)	$\langle e\rangle$	4

Department of Mathematics, Wetsern Washington University, Bellingham, WA 98225
Email address: jensenn5@wwu.edu

Department of Mathematics, Western Washington University, Bellingham, WA 98225
Email address: trenees@wwu.edu